

Multiplication Three Ways

Answer Key

Grade 3

Quick Answers

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|--------------------------------|----------------------------------|
| 1. how many 4s = 3, total = 12 | 7. how many 9s = 2, total = 18 |
| 2. how many 6s = 5, total = 30 | 8. total = 30, — |
| 3. total pages = 24, — | 9. total stars = 35, — |
| 4. total dots = 18, — | 10. how many 10s = 4, total = 40 |
| 5. total = 28, — | 11. 5 |
| 6. total dots = 20, — | 12. (c) = 54, — |
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What is verified

5 of 12 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problems 3, 4, 5, 6, 8, 9, 12. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 3

3.OA.A.1 – problems 1, 2, 4, 5, 6, 7, 8, 9, 10

3.OA.A.3 – problems 3, 12

3.OA.A.4 – problem 11

Difficulty 1–4 of 5 across 24 tagged checks.

Common wrong answers

3. *If they got 11: added the days and the pages per day*
12. *If they got 15: added the rows and the chairs in each row*
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1. Fill in: $4 + 4 + 4 = _ \times 4 = _$

Solution: The 4 repeats 3 times, so the fact is 3×4 . Adding gives $4 + 4 + 4 = 12$.

$$4 + 4 + 4 = 3 \times 4 = 12$$

2. Fill in: $6 + 6 + 6 + 6 + 6 = _ \times 6 = _$

Solution: The 6 repeats 5 times, so the fact is 5×6 . Skip counting by 6: 6, 12, 18, 24, 30.

$$6 + 6 + 6 + 6 + 6 = 5 \times 6 = 30$$

3. Mia reads 8 pages each day for 3 days. Multiplication fact and total?

Solution: Three days of 8 pages is 3 equal groups of 8, so the fact is $3 \times 8 = 24$ (or $8 \times 3 = 24$). The total is $8 + 8 + 8 = 24$ pages. 24

4. Array with 3 rows of 6 dots: multiplication fact and total.

Solution: 3 rows of 6 gives the fact $3 \times 6 = 18$ (or $6 \times 3 = 18$). Counting all dots: $6 + 6 + 6 = 18$. 18

5. Write 4×7 as repeated addition, then find the total.

Solution: 4×7 means 4 groups of 7: $7 + 7 + 7 + 7$. (Seven 4s added is also correct — it gives the same total.) Adding: $7 + 7 = 14$, $14 + 7 = 21$, $21 + 7 = 28$. 28

6. Ari's array: 5 rows with 4 dots in each row. Repeated addition and total.

Solution: One 4 for each of the 5 rows: $4 + 4 + 4 + 4 + 4$. (By columns, $5 + 5 + 5 + 5 + 5$ also shows the array.) Either way the total is $5 \times 4 = 20$ dots. 20

7. Fill in: $9 + 9 = _ \times 9 = _$

Solution: The 9 repeats 2 times, so $9 + 9 = 2 \times 9 = 18$.

$$9 + 9 = 2 \times 9 = 18$$

8. Write 6×5 as repeated addition, then find the total.

Solution: 6×5 means 6 groups of 5: $5 + 5 + 5 + 5 + 5 + 5$. (Five 6s added is also correct.) Skip counting by 5 six times: 5, 10, 15, 20, 25, 30. 30

9. Star array: 7 rows with 5 stars in each row. Multiplication fact and total.

Solution: 7 rows of 5 gives $7 \times 5 = 35$ (or $5 \times 7 = 35$). Skip counting by 5 seven times lands on 35. 35

10. Fill in: $10 + 10 + 10 + 10 = _ \times 10 = _$

Solution: The 10 repeats 4 times, so $10 + 10 + 10 + 10 = 4 \times 10 = 40$.

$$10 + 10 + 10 + 10 = 4 \times 10 = 40$$

11. How many 8s make 40? $8 \times n = 40$

Solution: Skip count by 8 until you reach 40: 8, 16, 24, 32, 40 — that is five 8s. So $8 \times 5 = 40$. $n = 5$

12. Challenge: 6 rows of chairs with 9 chairs in each row.

Solution (a): The array fact is $6 \times 9 = 54$ (or $9 \times 6 = 54$).

Solution (b): One 9 for each row: $9 + 9 + 9 + 9 + 9 + 9$ (six 9s; adding 6 nine times also works).

Solution (c): The total is $6 \times 9 = 54$ chairs — not $6 + 9 = 15$, which only adds the row count to one row's chairs. 54